

Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

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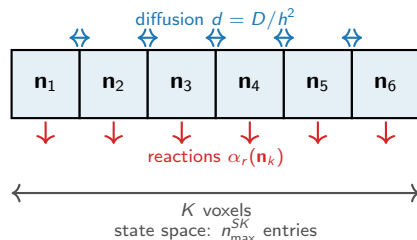
The RDME: reactions and diffusion across a spatial grid

Model. Divide a 1-D domain into K voxels. Each voxel k holds molecule counts $\mathbf{n}_k \in \mathbb{Z}_{\geq 0}^S$.

Two types of events:

- **Reactions** inside voxel k : rate $\alpha_r(\mathbf{n}_k)$, change $\mathbf{n}_k$.
- **Diffusion** between neighbors: rate $d = D/h^2$, moves one molecule $k \leftrightarrow k+1$.

The joint probability $p(\mathbf{n}_1, \dots, \mathbf{n}_K, t)$ satisfies $\dot{p} = Ap$ (**RDME**).



The computational bottleneck

The state space has $\sim n_{\max}^{SK}$ reachable states — **exponential in K** . Even $K=8$, $S=1$ is intractable with a naive FSP.

$\dot{p} = Ap$ has **block-tridiagonal structure** (schematic, $K=4$, $S=1$)

The diagram illustrates the block-tridiagonal structure of the matrix A and the corresponding vector equation $\dot{p} = Ap$.

The matrix A is a 4×4 block matrix with blocks A_k (blue) on the diagonal and $D_{k,k+1}$ (red) on the off-diagonals. The rows and columns are indexed by n_1, n_2, n_3, n_4 .

The vector equation is shown as:

$$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix} = \begin{bmatrix} \dot{p}_1 \\ \dot{p}_2 \\ \dot{p}_3 \\ \dot{p}_4 \end{bmatrix}$$

Each block is of size $n_{\max}^S \times n_{\max}^S$. The total size is $n_{\max}^{SK} \times n_{\max}^{SK}$.

A_k : reactions inside voxel k
voxel k

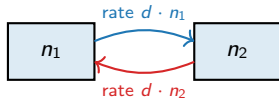
$D_{k,k+1}$: diffusion between neighboring voxels

p_k : probability sub-vector for

The Binomial is the exact diffusion equilibrium

Fix total molecules $\bar{n} = n_1 + n_2$ across two adjacent voxels.

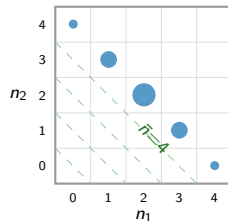
Diffusion moves one molecule at a time, at rate d per molecule:



Each molecule hops independently $\Rightarrow$ at equilibrium, equally likely to be in either voxel. For $\bar{n}$ total:

$$P(n_1 \mid \bar{n}) = \text{Bin}(\bar{n}, \frac{1}{2}) \quad (\text{exact, any } d > 0)$$

When the distribution is near this equilibrium, $\bar{n}$ **alone describes the two-voxel state.**



Probability spreads along constant-total diagonals as $\text{Bin}(\bar{n}, \frac{1}{2})$ (dot size $\propto$ mass).

The two-voxel state is captured by $\bar{n}$ alone.

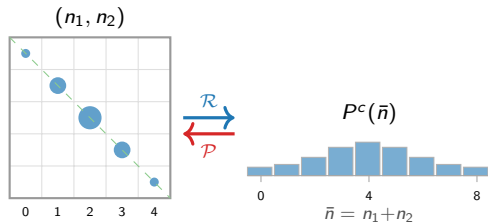
Merging two adjacent voxels: restriction and prolongation

Compress ($\mathcal{R}$): exact:

$$p^c(\bar{n}) = \sum_{n_1+n_2=\bar{n}} p(n_1, n_2).$$

Reconstruct ($\mathcal{P}$): approximate:

$$p(n_1, n_2) \approx p^c(\bar{n}) \cdot \text{Bin}(\bar{n}, \frac{1}{2}).$$



$\mathcal{R}$: sum along each diagonal $\rightarrow$ scalar $\bar{n} = n_1 + n_2$.

$\mathcal{P}$: spread back via $\text{Bin}(\bar{n}, \frac{1}{2})$.

State-space reduction

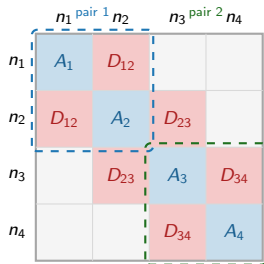
Per pair: $n_{\max}^{2S} \rightarrow n_{\max}^S$.

All K voxels: $n_{\max}^{SK} \rightarrow n_{\max}^{SK/2}$.

$\mathcal{P}$ is accurate when the distribution is near-Binomial :
verified at runtime, not assumed globally.

What merging does to the rate matrix

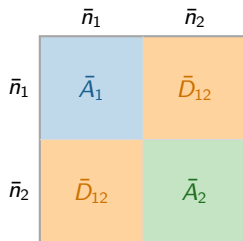
Full matrix A : 4 voxels:



Size: $n_{\max}^4$



Merged matrix $\bar{A}$: 2 pairs:



Size: $n_{\max}^2$ ($10^4 \times$ smaller for $n_{\max}=100$)

- $A_k \rightarrow \bar{A}_j$: reactions averaged over Binomial
- Within-pair D disappears: implicit in Binomial
- $\bar{D}_{12}$: only between-pair diffusion remains

The multigrid FSP algorithm

`multigrid_fsp( $u_0, T, \Delta t$ )`

❶ **Initialise:** $p \leftarrow \delta_{u_0}$; expand $\mathcal{S}$ by one shell

❷ **Detect front:**

$$\alpha_j = \frac{|\mu_{2j-1} - \mu_{2j}|}{\mu_{2j-1} + \mu_{2j}}, \quad j^* = \operatorname{argmax}_j \alpha_j$$

Keep pairs $j^* \pm 1$ fine; merge rest.

❸ **Restrict:** $\tilde{n}_j = n_{2j-1} + n_{2j}$ for merged pairs

❹ **Solve:** $p \leftarrow e^{\Delta t A^M} p$ (Krylov)

❺ **Prolong:** $\operatorname{Bin}(\tilde{n}_j, \frac{1}{2})$ for merged; dynamic- π at fine pairs

❻ **Expand:** add one-step reachable states to $\mathcal{S}$

❼ **Prune:** remove $\{x : p(x) < \tau\}$

❽ Repeat from ❷ until $t = T$

Cost per step

Coarse solve: $|\mathcal{S}^M| \ll N^K$

Expand: $O(|\mathcal{S}^M| \cdot |\text{reactions}|)$

Prune: $O(|\mathcal{S}^M|)$

$\mathcal{S}^M$ tracks only the support of p : never the full truncated space.

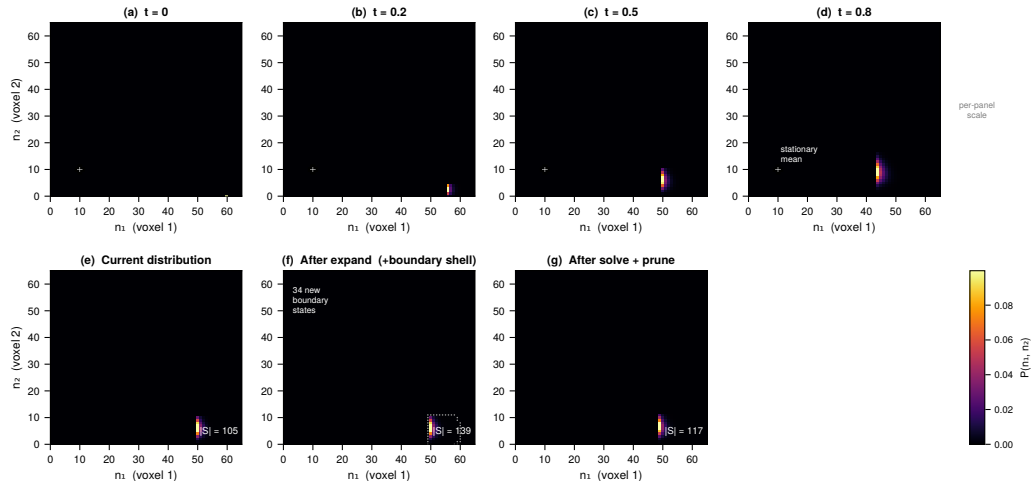
Key decision (step 2)

$\alpha_j \approx 0$: merge pair j (Binom exact)

$\alpha_j \approx 1$: keep fine (front present)

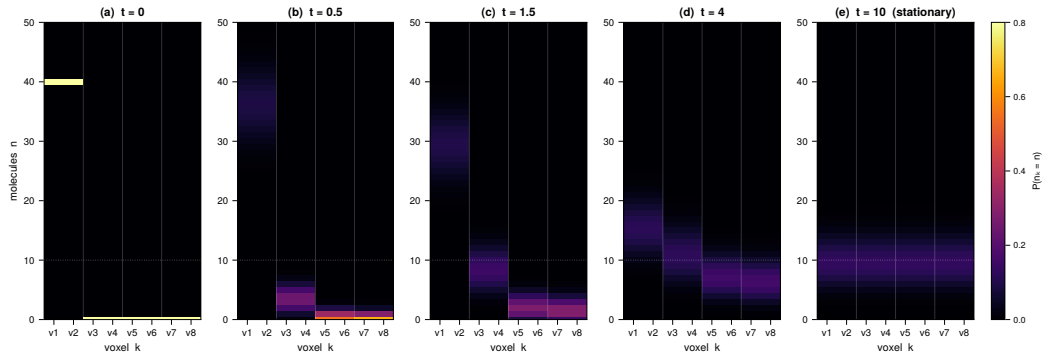
Same criterion for birth-death and Schlögl.

The adaptive FSP algorithm: expand, solve, prune



Top row: joint distribution $P(n_1, n_2)$ of a 2-voxel birth-death system: point source at $(n_{src}, 0)$ traveling to the stationary point (white cross).

Result: K=8 RDME wavefront via coarsened simulation



K=8 wavefront solved via K=4 coarse FSP. Max coarse state space: 106585 states. K=8 direct FSP would be intractable (grows to 90k+ states by $t=0.8$). Fine marginals recovered analytically via $\text{Binom}(n\tilde{n}, \frac{1}{2})$ prolongation — exact for balanced pairs.

Birth-death RDME, $K=8$, $D=0.1$. IC: $v_1=v_2=40$ (delta), $v_3 \dots v_8=0$. Each panel shows marginals $P(n_k=n)$ for all 8 voxels.

(a)–(d): Wavefront sweeps left→right, distributions broaden. **(e):** Stationary: all voxels reach $\text{Poisson}(10)$ (dashed line).

Solved via **K=4 coarse FSP** (30k states). Direct K=8 FSP: intractable. Fine marginals recovered via $\text{Bin}(\tilde{n}, \frac{1}{2})$:

What about bistable reactions?

Birth-death: coarsening is exact at stationarity.

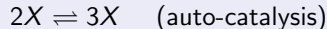
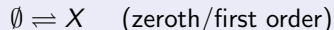
Each voxel has a single stable state. Pairs equilibrate to $\text{Bin}(\tilde{n}, \frac{1}{2})$: merging always valid.

But what if reactions are bistable?

Some systems have *two* stable states per voxel. Spatial fronts form where neighbors occupy opposite states. These fronts are metastable: persistent for very long times.

The Schlögl model

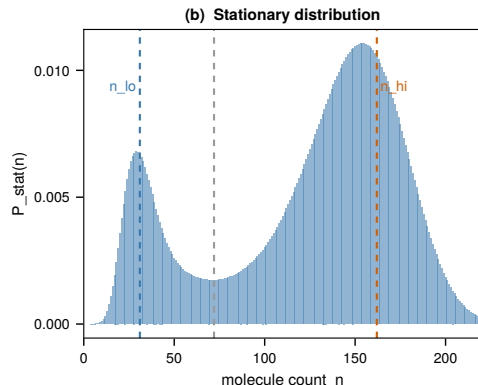
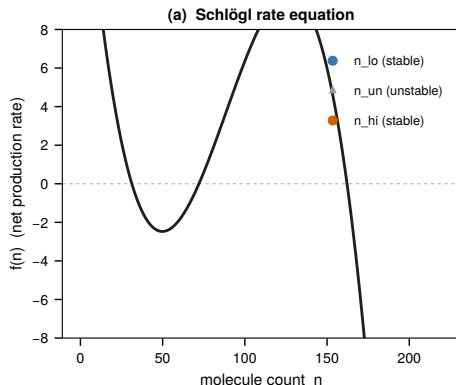
Bistable autocatalytic reactions:



Two stable states: $n_{\text{lo}} \approx 31$ and $n_{\text{hi}} \approx 162$.

At a spatial front: joint $P(n_k, n_{k+1})$ is *bimodal*
— Binomial prolongation is completely wrong.

Each voxel is bistable: two stable molecular states

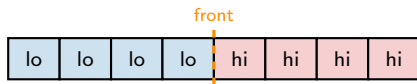


(a) Net production rate $f(n)$: three zeros: two stable (n_{lo} , n_{hi}), one unstable. **(b)** Stationary distribution is bimodal.

Think of it as a double-well potential: each voxel wants to be at n_{lo} or n_{hi} . Each spatial configuration: e.g., $(n_{lo}, n_{lo}, n_{hi}, n_{hi}, \dots)$: is metastable.

Bistable fronts: when does Binomial coarsening fail?

A **spatial front** forms when neighboring voxels occupy opposite stable states. Slow diffusion sustains it indefinitely.



At the interface pair: the joint $P(n_k, n_{k+1})$ is **bimodal** (two orientations: lo|hi and hi|lo).

Coarsening sums $\tilde{n} = n_k + n_{k+1}$: both orientations give the *same* total. Binomial prolongation puts both voxels at $\tilde{n}/2 \approx 97$.

Wrong.

The symmetry trap

$$(n_{\text{lo}}, n_{\text{hi}}) \xrightarrow{\mathcal{R}} \tilde{n} = 193 \xleftarrow{\mathcal{R}} (n_{\text{hi}}, n_{\text{lo}})$$

Both front orientations map to the same coarse state. $\mathcal{P}$ cannot recover the orientation.

Birth-death vs. Schlögl

	BD	Schlögl front
Within-pair dist.	Binomial	Bimodal
$\mathcal{P}$ accurate?	yes	no
α_j	≈ 0	≈ 1

When is merging valid? The imbalance criterion

Compare mean counts in adjacent voxels $2j-1$ and $2j$:

$$\alpha_j = \frac{|\mu_{2j-1} - \mu_{2j}|}{\mu_{2j-1} + \mu_{2j}}.$$

- $\alpha_j \approx 0$: voxels balanced $\Rightarrow$ distribution near-Binomial $\Rightarrow$ **safe to merge**.
- $\alpha_j \approx 1$: voxels in opposite states $\Rightarrow$ front is nearby $\Rightarrow$ **keep fine**.

Adaptive partitioning

Find $j^* = \operatorname{argmax}_j \alpha_j$ (front location).

Keep j^* and its immediate neighbors fine.

Merge all other pairs.

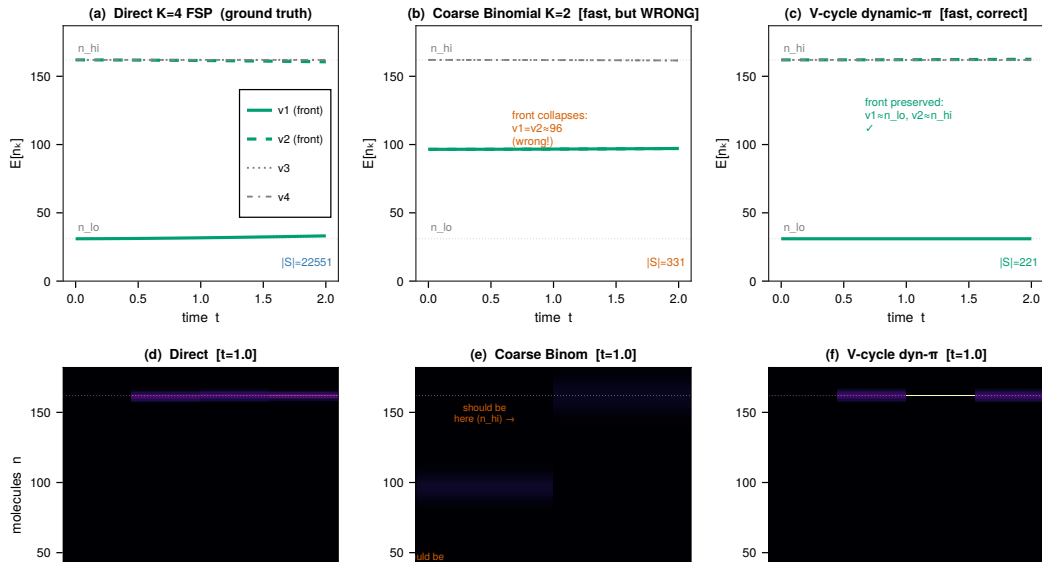
Recheck α_j only when the front moves.

One V-cycle step:

- 1 **Detect front** via α_j .
- 2 **Restrict** bulk pairs: $\bar{n}_j = n_{2j-1} + n_{2j}$.
- 3 **Coarse solve**: $\dot{p}^M = A^M p^M$ on $K/2$ effective voxels.
- 4 **Prolong** fine pairs back using dynamic- π (correct at front; Binomial elsewhere).

State space: $K/2$ coarse voxels + small fine window. Tractable where full K -voxel FSP is not.

Result: adaptive V-cycle correctly tracks the Schlögl front



Summary

1. Fast diffusion enables state-space compression.

When the joint distribution of two adjacent voxels is near $\text{Bin}(\bar{n}, \frac{1}{2})$, replacing (n_1, n_2) by $\bar{n}$ cuts the per-pair state space from $n_{\max}^{2S}$ to $n_{\max}^S$. *K=8 birth-death wavefront solved with 30k states (direct: intractable).*

2. Bistable fronts require dynamic- π prolongation.

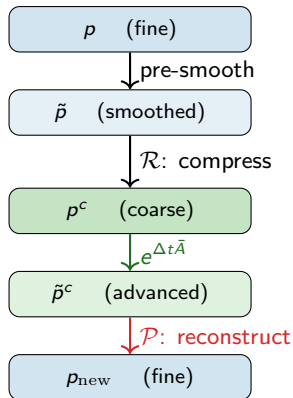
At a front, the within-pair distribution is bimodal: Binomial is wrong. The imbalance α_j detects this at runtime. V-cycle with dynamic- π corrects the prolongation. *K=4 Schlögl: V-cycle 221 states, direct 22k, both accurate.*

3. Same algorithm: α_j adapts automatically.

Birth-death: $\alpha_j \approx 0$ always $\Rightarrow$ merge all pairs. Schlögl: $\alpha_j \approx 1$ at front $\Rightarrow$ keep fine. One algorithm handles both.

Open: multi-level hierarchy; multi-species; moving fronts; rigorous error bounds.

Appendix: why not reconstruct at every step?

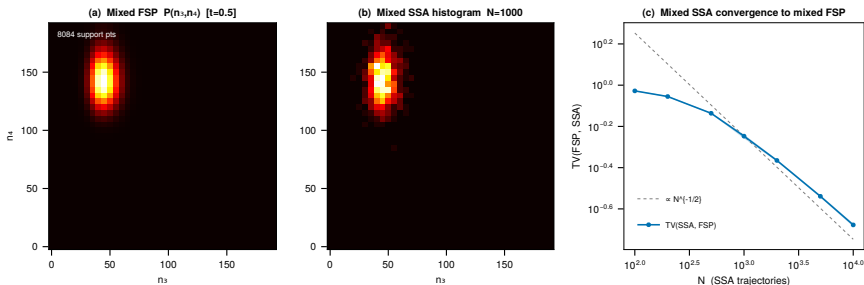


Reconstruction is wrong at the front

Step 4 forces a Binomial split on *every* pair, including the front where the true distribution is bimodal.

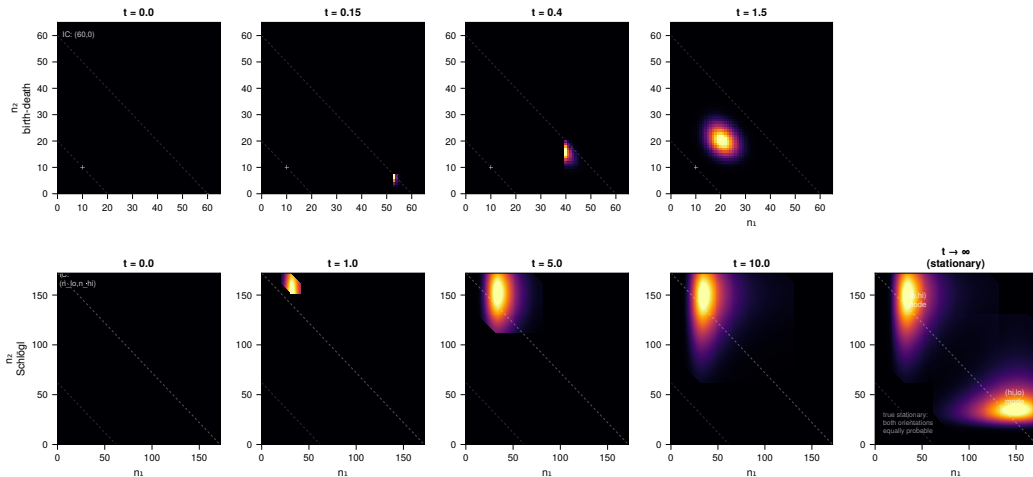
Fix: evolve p^M directly: no reconstruction step. Coarse pairs carry the Binomial assumption implicitly in $\bar{A}_j$; the fine window tracks the true within-pair distribution exactly.

Appendix: mixed-resolution generator is self-consistent



ODE solve on the mixed state $(\bar{n}_1, n_3, n_4)$ vs. Monte Carlo on the same reduced generator. Means agree to $<1\%$; $TV \sim N^{-1/2}$ (sampling noise only).

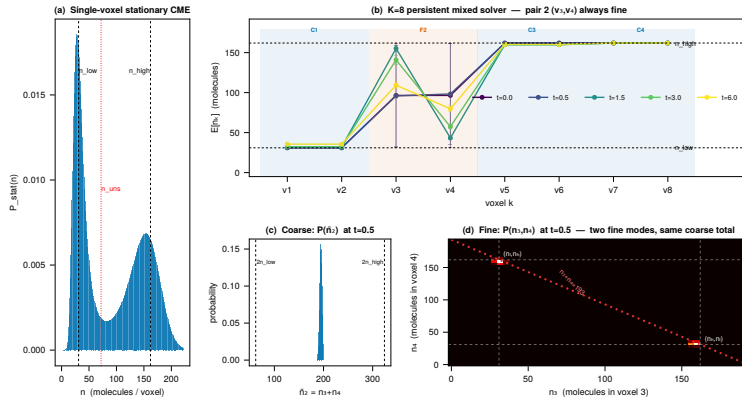
Appendix: probability dynamics (birth-death vs. Schlögl)



Top: birth-death point source traveling to Poisson(10) stationary. Binomial coarsening exact.

Bottom: Schlögl front IC persisting metastably; true stationary (rightmost) shows both orientations equally.

Appendix: K=8 persistent front (detailed)



Appendix: K=12 persistent front

